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Topological Classification of Immune Cell Spatial Patterns in the Tumor Microenvironment

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1 Introduction

Solid tumors are not merely collections of malignant cells: they are complex ecosystems in which the spatial arrangement of immune cells plays a decisive role in disease progression and patient prognosis (Vipond et al., 2021; Binnewies et al., 2018). In head and neck squamous cell carcinoma (HNSCC), the interplay between cytotoxic T cells (CD8), macrophages (CD68) and regulatory T cells (FoxP3) shapes the balance between immune attack and immune evasion. Characterising the *spatial organisation* of these populations beyond simple cell counts is a pressing question in computational pathology, and recent reviews highlight the rapid development of dedicated spatial-statistics and topological methods to address it (Heindl et al., 2015; Vipond et al., 2021).

Traditional spatial statistics (Ripley’s K , nearest-neighbour distributions, pair correlation functions) capture local geometric properties but struggle to summarise the global, multi-scale “shape” of a point cloud in a single descriptor. Topological Data Analysis (TDA), and in particular persistent homology, offers a complementary perspective: by tracking how topological features (connected components, loops, voids) appear and disappear across spatial scales, it produces a multi-resolution descriptor of spatial structure with provable stability properties (Chazal and Michel, 2021; Otter et al., 2017). Whether this richer descriptor brings added discriminative value over explicit coordinate-based geometric summaries in the tumor microenvironment is, however, an empirical question that is rarely examined directly within a single pipeline: TDA results on biological point clouds and classical spatial-statistics results typically appear in different studies, with no head-to-head comparison on the same data and the same evaluation protocol. A central goal of this report is to assess this comparison on a single dataset, while recognising that coordinate-based summaries can be computationally simple yet geometrically informative.

The Vipond et al. (2021) study. Our work builds directly on Vipond et al. (2021), who released a curated dataset of HNSCC immunohistochemistry images annotated with three immune cell types (CD8, CD68, FoxP3) and segmented into $1.5\text{ mm} \times 1.5\text{ mm}$ regions of interest (ROIs). They compute *multiparameter* persistent homology on each ROI, using a two-parameter filtration that combines the standard radius parameter with a *codensity* parameter acting as a proxy for local cell density (and indirectly for hypoxic status). Multiparameter persistence landscapes (Vipond et al., 2021) are then used as features in a classification pipeline, achieving up to 87% accuracy on pairwise cell-type discrimination tasks. The headline result is that multiparameter TDA discriminates immune cell types substantially better than unspecified one-parameter baselines. What the paper does *not* disentangle is the relative contribution of (i) topology itself versus explicit coordinate-based geometric summaries of the point clouds and (ii) the codensity parameter on top of standard one-parameter persistent homology.

Positioning of this work. Rather than reproducing the multiparameter pipeline, we use the same dataset to perform a focused *ablation study*: we evaluate, on the same biological problem and a rigorous evaluation protocol, what each component of a topological classifier actually contributes. We fix the cross-validation scheme to a tumor-level StratifiedGroupK-Fold, and we vary the feature representation along two axes: TDA versus classical spatial statistics, and one-parameter persistence (this work) versus multiparameter persistence (the Vipond benchmark). Our work plays a complementary role to Vipond et al. (2021): it provides the missing one-parameter baseline and the missing non-TDA baseline that put the multiparameter results in perspective, and it provides an empirical point of comparison for assessing how much performance may remain beyond the one-parameter setting.

Contributions. We:

1. Validate our pipeline on synthetic data with known topology (1D spheres in 2D), confirming that our implementation correctly recovers ground-truth H_0 and H_1 features and is robust to spatial noise.
2. Perform an exploratory analysis of two large hypoxic tumor regions to characterise the biological context and spatial architecture of immune infiltration.
3. Compute persistent homology (H_0 and H_1) for each cell type and vectorise the resulting persistence diagrams via persistence landscapes and persistence images.
4. Classify cell types using persistence-landscape features within a tumor-level cross-validation framework, comparing three classifiers with distinct inductive biases (regularised LDA, ℓ_2 -regularised logistic regression, gradient boosting), and explicitly discussing the assumptions that make each appropriate.
5. Conduct two-sample permutation tests on tumor-level mean landscape vectors and a paired within-tumor MMD test on the full distribution of tumor-level landscapes.
6. Compare the discriminative power of topological features against interpretable coordinate-based geometric summaries, and evaluate their combination, to empirically assess the marginal contribution of one-parameter TDA.

2 Literature Review

We organise the literature around the actual problem addressed in this report: classifying and statistically testing the spatial organisation of immune cells in tumor tissue. We first review non-TDA approaches to spatial pathology, then TDA approaches to the same problem, then the statistical machinery (vectorisations, hypothesis tests, classification) used to make inference on persistence diagrams possible.

2.1 Spatial Statistics for the Tumor Microenvironment

The idea that the spatial organisation of immune cells in a tumor carries prognostic information has been substantiated by a growing body of work in computational pathology (Heindl et al., 2015; Binnewies et al., 2018). The classical statistical tools used in this literature come from spatial point-process theory: Ripley’s K function and the closely related L and pair correlation functions (Ripley, 1976; Baddeley et al., 2015) quantify the deviation of a point pattern from complete spatial randomness across spatial scales; nearest-neighbour distance distributions (G and F functions) measure inter-cell proximity; mark correlation functions extend these tools to multi-type point processes. In oncology, these descriptors have been applied to quantify lymphocytic infiltration patterns and their association with patient outcome (Heindl et al., 2015; Maley et al., 2015), and to detect immune-exclusion zones around tumor cores. A recurring methodological question, also relevant to our work, is how to summarise the resulting curves into a fixed-dimensional feature vector amenable to standard supervised learning: typical choices include evaluating $K(r)$ at a small set of radii, integrating against weight functions, or fitting parametric point-process models. Reproducing an interpretable coordinate-based geometric baseline on the same dataset is a natural part of our ablation study, and motivates the features described in Section 4.6.

2.2 Topological Approaches to Spatial Biology

Persistent homology was developed in the early 2000s (Edelsbrunner et al., 2002; Zomorodian and Carlsson, 2005) to extract a multi-scale topological signature from a finite metric space: from a point cloud one builds a nested sequence of simplicial complexes (a *filtration*) by varying a scale parameter, and one tracks the appearance (*birth*) and disappearance (*death*) of topological features across the filtration. The output is a *persistence diagram*, a multiset of birth–death pairs (b, d) in the extended half-plane, in which features far from the diagonal are interpreted as topological signal and points close to the diagonal as noise. Otter et al. (2017) provide a now-standard computational roadmap for persistent homology, comparing the main filtrations (Vietoris–Rips, alpha, witness) and the available software pipelines; Chazal and Michel (2021) give a complementary statistically oriented introduction.

In the specific context of immune-cell spatial patterns, the most directly relevant work is Vipond et al. (2021). Their methodological choice is to move from one-parameter to *multiparameter* persistent homology: they construct a two-parameter filtration that varies both a Vietoris–Rips radius and a codensity parameter encoding local cell density, and they vectorise the resulting multiparameter persistence module via multiparameter persistence landscapes which they introduce in the same paper. The codensity acts as an implicit proxy for the hypoxic niche where regulatory T cells accumulate, and is presented as the main reason why their pipeline outperforms one-parameter baselines. Beyond this specific

application, persistent homology has been applied to a range of biological data: protein structure (Gameiro et al., 2015), brain connectivity (Chung et al., 2009) and single-cell transcriptomics (Rizvi et al., 2017). In all these settings, the central modelling question is the same as ours: which filtration, which homological dimensions and which vectorisation extract a signal that is both stable and discriminative, and how does this signal compare to explicit geometric descriptors.

Our work positions itself between the classical spatial-statistics literature and the multiparameter pipeline of Vipond et al. (2021): we use the same data and a comparable evaluation protocol but restrict ourselves to one-parameter persistent homology, in order to assess the contributions of (i) topology versus explicit coordinate-based geometric summaries and (ii) one-parameter versus multiparameter filtrations.

2.3 From Persistence Diagrams to Statistics: Vectorisations, Tests, Classifiers

Vectorisations. Persistence diagrams equipped with bottleneck or Wasserstein distances form a metric space with a non-Euclidean geometry, awkward for downstream statistics and machine learning. A now-standard remedy is to map diagrams into a Hilbert space via a stable vectorisation, after which all the usual tools (means, hypothesis tests, classifiers) become available. A comprehensive survey of these vectorisations is given by Ali et al. (2023). We use two of them, *persistence landscapes* (Bubenik, 2015) and *persistence images* (Adams et al., 2017); their precise definitions are given in Sections 4.2 and 4.3.

Hypothesis testing. Fasy et al. (2014) construct confidence sets for persistence diagrams by bounding which features can be attributed to sampling noise; this gives a non-parametric notion of “topological signal”. Robinson and Turner (2017) develop a permutation framework for two-sample comparisons of persistence diagrams directly in diagram space, using the bottleneck or Wasserstein metric. These works establish the broader question of statistical inference on persistence diagrams that our Section 4.4 addresses, although our concrete tests are different in nature: rather than working in diagram space (which we discuss but do not implement, for the reasons given in Section 4.4), we work in the Hilbert space of persistence landscapes, which makes classical permutation and kernel tests directly applicable.

Classification with topological features. Once a stable vectorisation is available, supervised learning on persistence-based features reduces to a standard multivariate problem. The literature has explored linear models (LDA, logistic regression) for their interpretability and tractable inference, kernel methods exploiting persistence-specific kernels, and tree-based ensembles for their ability to capture interactions among landscape coefficients. The

practical concern that drives our methodology is not the choice of classifier per se but the evaluation protocol: when several ROIs come from the same tumor, naive random splits leak tumor identity into the training set and overestimate generalisation. We address this by a tumor-level StratifiedGroupKFold cross-validation, which is the standard remedy in the digital-pathology literature whenever multiple regions come from the same patient.

Position of our work. Our pipeline borrows its data and biological framing from [Vipond et al. \(2021\)](#), its classical-spatial-statistics baselines from the spatial-pathology and point-process literature, its vectorisations from [Bubenik \(2015\)](#); [Adams et al. \(2017\)](#), and its tumor-level evaluation protocol from the standard practice in digital pathology. The originality of the contribution lies in their combination into an ablation study that quantifies the marginal value of one-parameter TDA over classical spatial statistics on the published Vipond dataset.

3 Data and Descriptive Statistics

3.1 Data Description

We work with two datasets from [Vipond et al. \(2021\)](#), both derived from immunohistochemistry images of HNSCC specimens. **Dataset 1: Large Hypoxic Regions (LHR).** Two large tissue regions (LHR1 and LHR2) in which each cell is described by its spatial coordinates (x, y) , its immune cell type (CD8, CD68 or FoxP3) and binary markers for hypoxia (CAIX+), tumor cells (PanCK+) and necrotic tissue (Necrosis). These regions serve as the basis for the exploratory analysis. **Dataset 2: 1.5 mm Regions of Interest (ROIs).** Non-overlapping $1.5 \text{ mm} \times 1.5 \text{ mm}$ regions per cell type, spanning 16 distinct tumors (labelled T_A through T_P). Each CSV file contains only the (x, y) coordinates of cells of a single type within one ROI. We discard ROIs with fewer than 20 cells, as such sparse point clouds do not produce meaningful topological features; this leaves 530 CD8, 502 FoxP3 and 549 CD68 ROIs (1581 in total). No hypoxia markers are available in this dataset. Dataset 2 is used for classification and statistical testing.

The three immune cell types play complementary roles in the anti-tumor immune response. CD8^+ T cells are cytotoxic lymphocytes that directly kill malignant cells; high spatial infiltration of CD8 within the tumor core is generally associated with a favourable prognosis. CD68^+ cells are macrophages that phagocytose cellular debris. FoxP3^+ cells are regulatory T cells that suppress the immune response, a role generally associated with poor prognosis in oncology.

3.2 Cell Composition, Hypoxia, and Spatial Architecture

The distribution of cell types is similar across LHR1 and LHR2 (Figure 1): CD8 and CD68 are present in roughly equal numbers, while FoxP3 cells are approximately half as numerous, suggesting a partially maintained anti-tumor response with a balanced presence of effector cells and a smaller regulatory population. Analysis of the CAIX+ marker (Figure 2) reveals a striking contrast: CD8 and CD68 cells are overwhelmingly normoxic, while FoxP3 cells exhibit a significant proportion of hypoxic cells. This pattern, reproducible across both regions, is consistent with established biology: hypoxia promotes the recruitment and accumulation of immunosuppressive regulatory T cells while creating a hostile environment for cytotoxic cells.

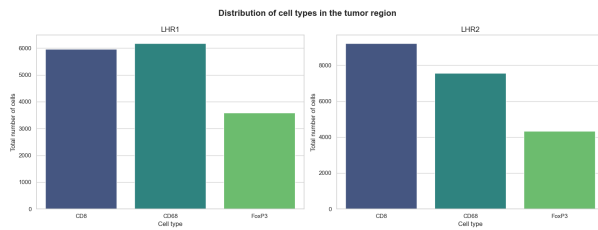


Figure 1: Distribution of immune cell types in LHR1 and LHR2.

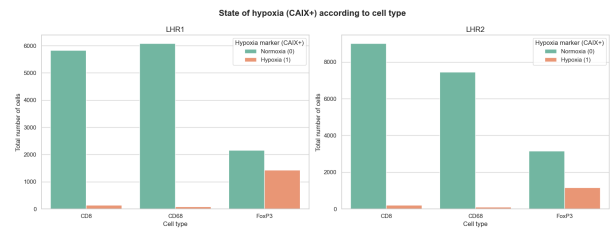


Figure 2: Hypoxic status (CAIX+) by cell type in LHR1 and LHR2.

The spatial maps (Figure 3) reveal a concentric architecture in both regions: a necrotic core surrounded by a hypoxic belt rich in FoxP3 cells, itself surrounded by a well-oxygenated periphery where CD8 and CD68 predominate. This layered organisation demonstrates how the physical and metabolic structure of the tumor shapes the distribution of immune cells, creating an immunosuppressive barrier that protects the tumor interior. This observation provides the biological motivation for our topological analysis: the distinct spatial configurations of different cell types should produce different topological signatures.

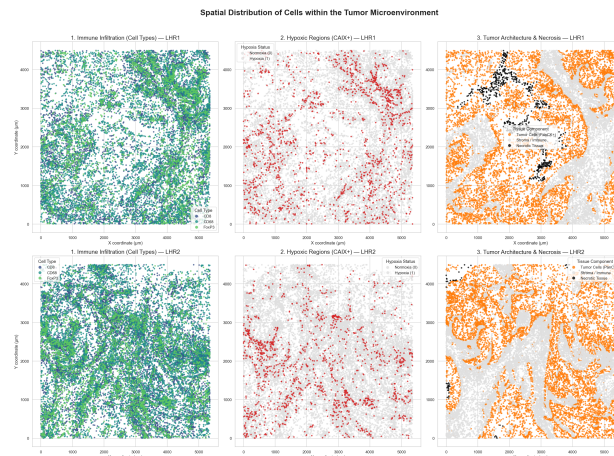


Figure 3: Spatial distribution of cells within the tumor microenvironment. Left: immune infiltration by cell type. Centre: hypoxic regions (CAIX+). Right: tumor architecture and necrosis. Top row: LHR1; bottom row: LHR2.

4 Methodology

4.1 Persistent Homology

Given a set of n points $\mathcal{P} = \{p_1, \dots, p_n\} \subset \mathbb{R}^2$ representing cell coordinates, we construct an **Alpha complex** filtration. The Alpha complex is the nerve of the Voronoi decomposition restricted to balls of radius $\sqrt{\alpha}$ centred at each point, and is particularly efficient in low dimensions. As the scale parameter α increases from 0, simplices are added to the complex, and topological features appear (are “born”) and disappear (“die”). We track features in two homological dimensions: H_0 (connected components) captures the clustering structure—a component is born when a point enters the filtration and dies when it merges with an older component; H_1 (loops) captures circular voids in the spatial distribution. The output is a **persistence diagram** $D = \{(b_i, d_i)\}_{i=1}^N$, where features far from the diagonal $\{b = d\}$ are persistent and considered topologically significant.

We use the `gudhi` library (GUDHI, 2014) for all persistent homology computations. Alpha complex filtration values correspond to squared radii; we apply a square-root transformation to recover Euclidean-scale birth/death coordinates. For the exploratory analysis, persistence images and landscapes are computed with the `persim` library; for the classification pipeline, we use `gudhi.representations.Landscape` so that the landscape transformer can be fitted train-side only inside scikit-learn pipelines.

4.2 Persistence Landscapes

Following Bubenik (2015), we vectorise each persistence diagram using persistence landscapes. For a diagram $D = \{(b_i, d_i)\}$, each point generates a “tent” function $f_{(b,d)}(t) = \max(0, \min(t - b, d - t))$. The k -th persistence landscape $\lambda_k : \mathbb{R} \rightarrow \mathbb{R}$ is the k -th largest value among all tent functions at each point t . We discretise each landscape on a grid of 80 equispaced points and retain the first 10 landscape depths ($k = 1, \dots, 10$), yielding a vector of dimension $80 \times 10 = 800$ per homological dimension. When combining H_0 and H_1 , the total feature vector has dimension 1600. The landscape representation is computed using the `gudhi.representations.Landscape` transformer, which integrates seamlessly into scikit-learn pipelines.

4.3 Persistence Images

For the exploratory analysis of the large hypoxic regions, we also compute persistence images (Adams et al., 2017). Each diagram point (b, d) is mapped to the birth-persistence plane $(b, d - b)$, and a weighted Gaussian kernel is placed at each transformed point. The resulting surface is discretised on a shared grid to produce a fixed-size matrix. We use a linear weighting function that gives more importance to high-persistence features.

4.4 Statistical Testing for Topological Differences

The central research question of this report is whether the spatial organisation of immune cell types—captured by their topological signatures—differs significantly. Several distinct null hypotheses can be entertained: do the *mean* topological summaries differ? do the *full distributions* of topological summaries differ? These two questions probe progressively stronger forms of difference and require distinct test statistics.

1. Do the mean topological summaries differ? As a primary test, we assess whether there is a global shift in the mean topological signature. We compute the mean H_1 landscape vector for each cell type at the tumor level (one vector per tumor and per cell type, averaging over the ROIs of that cell type within that tumor). The test statistic is the L^2 distance between the group means of these tumor-level vectors. Under the null hypothesis, the two cell-type labels are exchangeable within each tumor. We therefore use a paired within-tumor sign-flip permutation test: for each tumor, we independently swap or keep the two cell-type labels, and recompute the L^2 distance between the two group mean landscape vectors. This preserves the paired structure of the data and accounts for the fact that the same tumors contribute observations for all cell types. We use 10,000 such permutations and a +1 smoothing correction for the empirical p-value.

2. Do the full distributions of topological summaries differ? Two cell types might have identical mean landscapes but very different variances or distributional shapes, which the previous test cannot detect. To test whether the full distributions of topological summaries differ, we use a paired **Maximum Mean Discrepancy** (MMD) test. For each tumor that has ROIs of both cell types we form a paired observation $(\bar{\lambda}_A^{(t)}, \bar{\lambda}_B^{(t)})$, where $\bar{\lambda}_A^{(t)}$ is the mean H_1 landscape of type A in tumor t . The test statistic is MMD^2 with an RBF kernel between the two paired sets. The exchangeable null “ A and B produce the same distribution of mean landscapes within a tumor” is enforced by a within-tumor sign-flip permutation: for each tumor independently, with probability $1/2$ we swap its A/B labels. This respects the paired structure of the data and is the correct exchangeability assumption when ROIs are nested within tumors. We use 10,000 sign-flip permutations and a +1 smoothing on the empirical p -value.

Multiple comparisons. For all three pairwise comparisons we apply a Bonferroni correction: a result is declared significant only if its p -value is below $\alpha^* = 0.05/3 \approx 0.0167$.

4.5 Classification Pipeline

We evaluate whether topological signatures can distinguish immune cell types using the dataset of 1581 ROIs. The pipeline proceeds as follows: (i) for each ROI, we compute H_0 and H_1 persistence diagrams using the Alpha complex filtration on the full point cloud (no

subsampling); (ii) persistence diagrams are transformed into landscape vectors (10 depths, 80 grid points per depth), with the **Landscape** transformer fitted on the training fold only; (iii) all features are centred and scaled using a **StandardScaler** fitted on the training fold; (iv) we evaluate three classifiers with distinct inductive biases.

Classifiers and assumptions. The three classifiers are chosen to span a spectrum of inductive biases, since the geometry of persistence-landscape feature space is not known a priori.

Regularised LDA. Linear Discriminant Analysis (LDA) is the classical linear classifier derived from a Gaussian generative model with a shared within-class covariance matrix. Both Gaussianity and shared covariance are violated to some extent in our setting (landscape coefficients are non-negative and skewed, and there is no biological reason for the three cell types to share covariance), so we do not interpret LDA as a generative model. We use it instead as a **regularised linear baseline**: even when its generative assumptions fail, LDA still produces a valid linear decision boundary, and comparing it to logistic regression—another linear classifier with different inductive biases—provides an indirect check that the linear part of the signal is captured consistently. The technical issue is that the empirical within-class covariance matrix has size $d \times d$ with $d \in \{800, 1600\}$ and $n \approx 1500$ ROIs, which is severely ill-conditioned and would make LDA numerically unstable. We therefore use sklearn’s `LinearDiscriminantAnalysis(solver='lsqr', shrinkage='auto')`, which applies **Ledoit–Wolf shrinkage**: the empirical covariance $\hat{\Sigma}$ is replaced by $(1 - \lambda)\hat{\Sigma} + \lambda \frac{\text{tr}(\hat{\Sigma})}{d} I$, where λ is chosen automatically to minimise expected squared error to the true covariance. This regularises the estimator towards a scaled identity and makes LDA stable in the high-dimensional, multicollinear regime imposed by discretised landscape vectors.

ℓ_2 -regularised logistic regression. Logistic regression makes no distributional assumption on the features and only requires the conditional log-odds to be linear in the inputs. We use ℓ_2 regularisation (default `C=1`) to control multicollinearity and the lbfgs solver with up to 1 000 iterations. As above, regularisation is essential because adjacent landscape grid points are nearly collinear.

Gradient boosting. A non-parametric ensemble of shallow regression trees that can capture non-linear interactions among landscape coefficients without geometric assumptions. We initially used an arbitrary configuration (`n_estimators=200, max_depth=3`) as a working choice, which yielded a 98.7% training accuracy versus a 60.1% test accuracy (+38.6 pp gap), a clear sign of overfitting that motivated a principled tuning step. We tuned (`n_estimators, max_depth`) on the 3-way $H_0 + H_1$ task with an inner 5-fold `StratifiedGroupKFold` (grouped by tumor) over `n_estimators` $\in \{50, 100, 200, 300\}$ and `max_depth` $\in \{2, 3, 4\}$. Selecting the configuration with the highest mean inner-CV ac-

curacy tends to favour the most flexible models in a grid: in our case, the highest-CV configuration would have given a near-perfect training accuracy with an unchanged test accuracy and an even larger train/test gap, exactly the failure mode we wanted to avoid. We therefore apply the *one-standard-error rule* of Breiman et al. (1984): among configurations whose mean CV accuracy is within one standard error of the best, we select the simplest (lowest $\text{n_estimators} \times 2^{\text{max_depth}}$). On our grid, the best inner-CV accuracy was 59.2% with a one-SE threshold of 56.6%, which left 12 configurations within tolerance; the simplest of these is ($\text{n_estimators}=50, \text{max_depth}=2$). We then fixed this conservative configuration for all subsequent Gradient Boosting experiments. Re-evaluating this fixed model with the 10-fold tumor-level protocol yields 74.9% training accuracy and 60.7% test accuracy (+14.2 pp gap), shrinking the train/test gap by more than half compared to the initial arbitrary configuration while slightly improving the test accuracy. This tuning step was used primarily to avoid a clearly overfitting Gradient Boosting specification, and should not be interpreted as a fully nested hyperparameter-optimised performance estimate. All gradient boosting results below should therefore be read as performance under the fixed selected configuration. While a formal nested cross-validation would theoretically eliminate any risk of data leakage, fixing a single, highly conservative, low-capacity configuration (i.e., a very shallow forest with $\text{n_estimators}=50$ and $\text{max_depth}=2$) heavily mitigates this risk. Our methodological goal was not to aggressively optimize hyperparameters to maximize accuracy, but rather to establish a stable, non-overfitting non-linear baseline to fairly contrast with linear models (LDA and Logistic Regression).

Justification of the model panel. The progression LDA $\rightarrow$ Logistic Regression $\rightarrow$ Gradient Boosting is by increasing model flexibility and decreasing interpretability, which provides a principled way to attribute performance gains to specific model capacities (Gaussian assumptions, linear decision boundary, non-linear interactions). If linear models (LDA, LogReg) reach the same accuracy as gradient boosting, the discriminative information is essentially linear in landscape features; if gradient boosting wins by a margin, non-linear interactions matter.

Cross-validation protocol. Because multiple ROIs come from the same tumor, a standard random split places ROIs from the same patient in both training and test sets, which inflates accuracy. We use **StratifiedGroupKFold** with 10 folds, where the grouping variable is the tumor identifier. This ensures all ROIs from a given tumor appear in the same fold, providing a tumor-level generalisation estimate. The **Landscape** transformer and the **StandardScaler** are fitted on the training fold only inside each split.

4.6 Coordinate-Based Baseline Features

To assess whether persistent homology provides information beyond explicit coordinate-based geometric summaries, we construct a 15-dimensional feature vector from the raw cell coordinates, after centring and scaling each point cloud to unit RMS radius. The features summarise the relative geometry of the point cloud through four complementary axes: *spread and anisotropy* (4 features: standard deviation and range of x and y); *nearest-neighbour distances* (5 features: mean, standard deviation, minimum, maximum, median); *pairwise distances* (5 features: mean, standard deviation, median, 25th and 75th percentiles); and *convex hull area* (1 feature). Count is not included as an explicit feature, and each point cloud is centred and rescaled to unit RMS radius. However, nearest-neighbour distances, pairwise-distance summaries and convex-hull geometry can still be indirectly sensitive to the number and local density of points.

The choice of these features deserves a comment. They are not strictly independent of TDA: pairwise and nearest-neighbour distance statistics summarise the same point-cloud geometry that persistent homology captures, and the convex-hull area encodes a global geometric extent that one-parameter persistence reflects only indirectly. Although these features are simple to compute and interpret, they should not be viewed as a weak or naive baseline: they directly summarise spread, anisotropy, nearest-neighbour distances, pairwise distances and convex-hull geometry. If these explicit geometric summaries already capture predictive information comparable to one-parameter persistence landscapes, then the marginal value of this particular TDA representation is limited on this dataset, and the added value reported by [Vipond et al. \(2021\)](#) may plausibly come from the codensity parameter.

4.7 Methodological Limitations

While our pipeline is rigorous in preventing data leakage, we acknowledge certain statistical limitations. To strictly avoid pseudoreplication, our tests and cross-validation outer loops were aggregated at the tumor level, resulting in a small effective sample size ($N = 16$). Consequently, our 10-fold Stratified GroupKFold leaves only 1 or 2 tumors in the test set per fold, generating high inter-fold variance. The lack of statistical significance in the CD8 vs. FoxP3 topological comparison (via MMD) must therefore be interpreted cautiously: it is highly likely driven by a lack of statistical power ($N = 16$) rather than true topological equivalence.

5 Results

5.1 Validation on Synthetic Data

Before applying our pipeline to the real tumor data, we validated it on synthetic data with known topology. We generated 1-dimensional spheres (circles) embedded in 2D using the `tadasets` library, in two regimes: a “clean” distribution (points sampled exactly on the circle) and a “noisy” distribution (additive Gaussian spatial noise, $\sigma = 0.2$). The ground-truth topology is unambiguous: exactly one prominent H_0 feature (one connected component) and one prominent H_1 feature (the circular void).

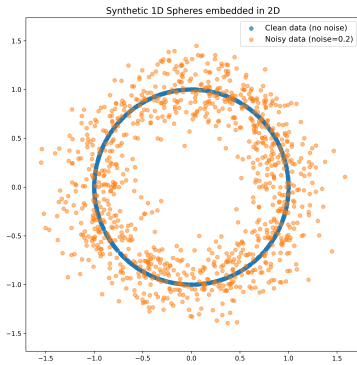


Figure 4: Synthetic 1D spheres in 2D, clean and noisy.

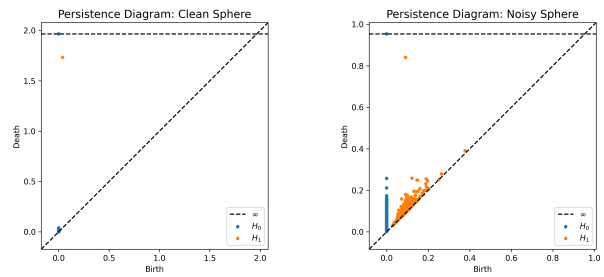


Figure 5: Persistence diagrams for the clean (left) and noisy (right) spheres. The prominent H_1 feature remains far from the diagonal in both cases.

The persistence diagrams (Figure 5) correctly capture the expected topology of the synthetic circles: a single highly persistent H_1 feature appears far from the diagonal in both the clean and noisy regimes. When noise is added, additional short-lived features appear near the diagonal, while the main loop remains visible. This preliminary experiment provides a useful sanity check that the main computational steps behave as expected on a controlled toy problem before applying the pipeline to the real immunohistochemistry data. A broader exploratory phase on synthetic shapes, including persistence diagrams, landscapes and images, was carried out in our first progress note and helped guide the implementation choices used here.

5.2 Exploratory Topological Analysis of the Large Hypoxic Regions

For computational tractability, the exploratory topological summaries of the large hypoxic regions were computed on the first 1,500 cells of each cell type and region rather than on the full point clouds. This deterministic truncation affects the persistence diagrams in Figure 6, as well as the persistence images and landscapes derived from them in Figures 7

and 8; it may therefore be sensitive to the ordering of cells in the input files. The spatial maps in Figure 3, however, use all cells from LHR1 and LHR2.

The persistence diagrams in Figure 6 show that H_0 features for all three cell types are concentrated near the vertical axis (birth ≈ 0), indicating that each cell population forms a densely connected distribution at the merging scale. The H_1 features span a similar range of birth and death values across the three cell types, with high-persistence loops visible in all configurations. Differences between cell types are therefore mainly quantitative—in the density of features and in slight shifts of the persistence range—rather than visually categorical. These exploratory observations motivate the more discriminative vectorisations and formal statistical tests used in the next sections.

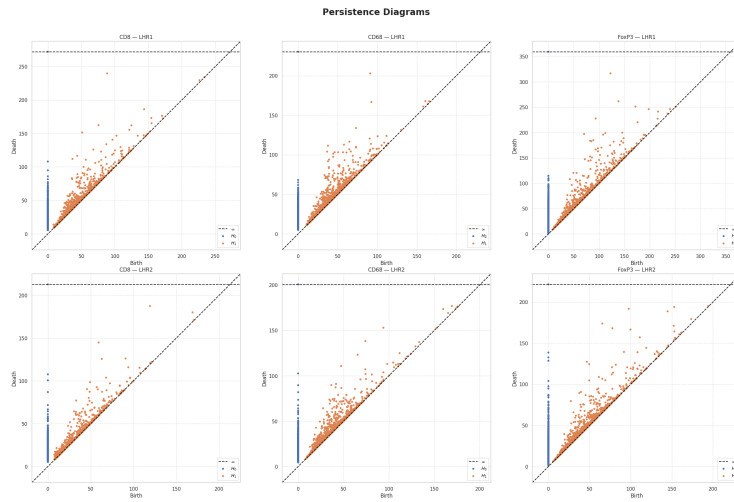


Figure 6: Persistence diagrams (H_0 and H_1) for each cell type in LHR1 and LHR2. H_0 features lie along the vertical axis for all three cell types; H_1 features span a comparable range across cell types, with quantitative rather than qualitative differences.

The persistence images (Figure 7) confirm this: all three cell types share a similar overall pattern, with a high-density region at low birth and low persistence and a few isolated high-persistence pixels. The persistence landscapes (Figure 8) show similar tent-shaped curves across cell types, with variation in amplitude across landscape depths but no obvious cell-type-specific signature visible to the naked eye. Quantitative differences between cell types are best assessed through the classification and statistical tests of Section 5 rather than through visual inspection of these exploratory plots.

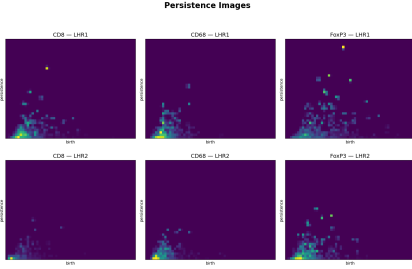


Figure 7: Persistence images (H_1) by cell type and region.

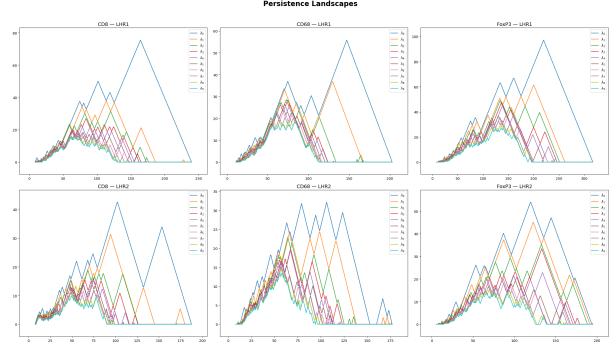


Figure 8: Persistence landscapes (H_1) by cell type and region.

5.3 Classification with One-Parameter Persistence Landscapes

5.3.1 Landscape Features Only

Table 1: Mean accuracy ($\pm$ std) on landscape features, by classifier and feature configuration (10-fold tumor-level StratifiedGroupKFold). Standard deviations across folds are typically in the 7–15 pp range, reflecting tumor-to-tumor heterogeneity.

Task	Classifier	H1 only	H0 only	H0 + H1
CD8 vs FoxP3	LDA	72.2% $\pm$ 10.3%	63.1% $\pm$ 14.9%	71.4% $\pm$ 10.5%
	Logistic Reg.	71.3% $\pm$ 9.0%	61.0% $\pm$ 12.8%	71.6% $\pm$ 8.1%
	Grad. Boosting	77.1% $\pm$ 8.7%	63.7% $\pm$ 15.3%	75.9% $\pm$ 9.0%
CD8 vs CD68	LDA	63.9% $\pm$ 9.7%	62.8% $\pm$ 7.7%	64.4% $\pm$ 7.9%
	Logistic Reg.	63.2% $\pm$ 8.9%	60.7% $\pm$ 6.2%	62.7% $\pm$ 7.6%
	Grad. Boosting	66.6% $\pm$ 11.5%	63.9% $\pm$ 7.5%	66.0% $\pm$ 10.8%
FoxP3 vs CD68	LDA	75.2% $\pm$ 11.0%	67.9% $\pm$ 16.4%	71.7% $\pm$ 13.5%
	Logistic Reg.	74.3% $\pm$ 7.3%	66.8% $\pm$ 15.7%	69.9% $\pm$ 12.4%
	Grad. Boosting	78.3% $\pm$ 12.3%	67.5% $\pm$ 16.4%	77.7% $\pm$ 12.7%
3-way	LDA	56.3% $\pm$ 9.9%	52.1% $\pm$ 11.0%	55.4% $\pm$ 9.3%
	Logistic Reg.	52.5% $\pm$ 7.0%	49.8% $\pm$ 11.1%	52.1% $\pm$ 7.8%
	Grad. Boosting	59.9% $\pm$ 10.0%	51.5% $\pm$ 11.9%	60.7% $\pm$ 10.1%

Four patterns emerge from Table 1.

(i) H_1 carries more discriminative signal than H_0 . Across all 12 (task, classifier) combinations, the H_0 -only column has the lowest accuracy: it falls to 49.8%–67.9%, with the 3-way accuracy dropping to 51.5%–52.1%. In contrast, H_1 -only reaches 63.2%–78.3% on pairwise tasks and up to 59.9% on the 3-way task. Most of the discriminative content of the landscapes therefore sits in the loop-scale features.

(ii) Adding H_0 to H_1 does not improve, and sometimes hurts, performance. For 10 of the 12 (task, classifier) combinations, $H_0 + H_1$ accuracy is within ± 2 pp of H_1 -

only. The two exceptions are FoxP3 vs CD68 with the linear classifiers, where $H_0 + H_1$ is 3.5 pp below H_1 -only for LDA ($75.2\% \rightarrow 71.7\%$) and 4.4 pp below for logistic regression ($74.3\% \rightarrow 69.9\%$); gradient boosting on the same task is essentially unaffected ($78.3\% \rightarrow 77.7\%$). This is consistent with the bias–variance trade-off for linear classifiers in high dimension: doubling the feature space by adding a weakly informative H_0 block can dilute the linear signal even when the strongest signal in H_1 is preserved.

(iii) FoxP3-containing pairs are easier than CD8 vs CD68. The two FoxP3-containing pairwise tasks reach the highest accuracies under every classifier: gradient boosting on H_1 achieves 77.1% on CD8 vs FoxP3 and 78.3% on FoxP3 vs CD68, against 66.6% on CD8 vs CD68. The CD8 vs CD68 pair is bounded between 60.7% and 66.6% across all configurations, regardless of classifier or homological dimension. This is consistent with the biological picture suggested by the exploratory LHR analysis, where FoxP3 cells appear more associated with hypoxic regions, although hypoxia markers are not available in the ROI dataset used for classification.

(iv) Non-linear modelling helps when the signal is in H_1 . On configurations involving H_1 (H_1 -only or $H_0 + H_1$), gradient boosting outperforms the average of LDA and logistic regression by +3.0 to +7.0 pp across the four tasks. On H_0 -only, the gain from gradient boosting drops to +0.2 to +2.1 pp. Gradient boosting therefore appears to exploit non-linear interactions among H_1 landscape coefficients that the linear classifiers cannot capture, but only when the underlying signal is strong enough; on the weak H_0 -only signal, non-linear modelling does not help. LDA and logistic regression yield very similar accuracies in every cell of the table (within ~ 1 – 2 pp), suggesting that the linear decision boundary is consistent under both inductive biases.

Standard deviations of 7–15 pp across the ten folds are large and similar across classifiers, so most of the variability in performance comes from tumor-to-tumor heterogeneity rather than from classifier choice. With only ~ 2 test tumors per fold, the biological variability between tumors is an intrinsic limit on the achievable per-fold accuracy.

Figure 9 shows the confusion matrices on $H_0 + H_1$ landscape features, pooled over the 10 folds. The most visible pattern is shared by the three classifiers: the largest off-diagonal mass falls on the CD8/CD68 pair, in both directions, which matches the lower pairwise accuracy on CD8 vs CD68 in Table 1. The FoxP3 row and column carry the smallest off-diagonal mass, consistent with the higher pairwise accuracies on the two FoxP3-containing tasks. Differences between the three classifiers in the confusion patterns are visually subtle and consistent with the small accuracy ranking (60.7% for gradient boosting vs 55.4% for LDA vs 52.1% for logistic regression on the 3-way task).

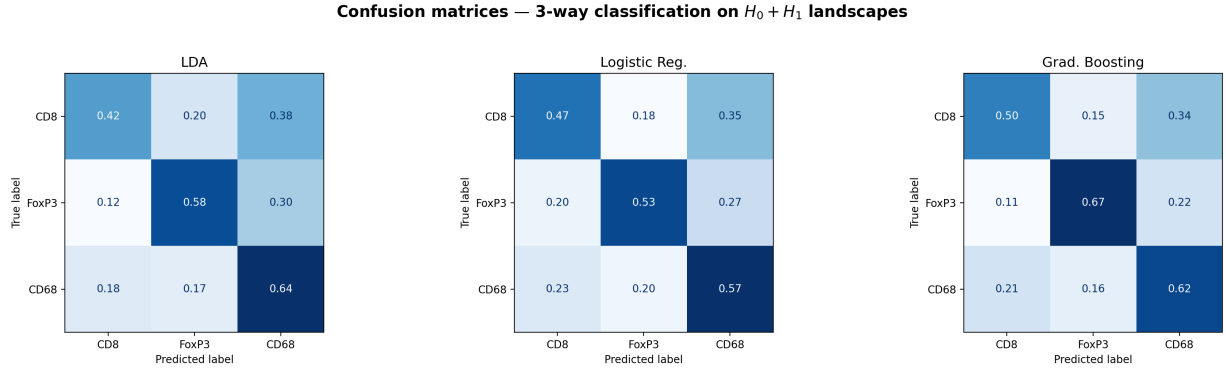


Figure 9: Confusion matrices (row-normalised) of the three classifiers on the 3-way task, using $H_0 + H_1$ landscape features and 10-fold tumor-level StratifiedGroupKFold. From left to right: regularised LDA, logistic regression, gradient boosting. The largest off-diagonal mass for every classifier falls on the CD8/CD68 pair.

5.3.2 Comparison with Coordinate-Based Features

Table 2 compares coordinate-based features versus H_1 persistence landscapes, and Table 3 reports the effect of combining $H_0 + H_1$ landscapes with coordinate features (gradient boosting only).

Table 2: Mean accuracy of coordinate features vs. H_1 landscapes, by classifier (10-fold tumor-level StratifiedGroupKFold).

Task	LDA		Logistic Reg.		Grad. Boosting	
	Coord	Landsc. (H_1)	Coord	Landsc. (H_1)	Coord	Landsc. (H_1)
CD8 vs FoxP3	79.4%	72.2%	90.2%	71.3%	92.2%	77.1%
CD8 vs CD68	94.8%	63.9%	94.1%	63.2%	95.4%	66.6%
FoxP3 vs CD68	80.7%	75.2%	86.7%	74.3%	91.5%	78.3%
3-way	78.1%	56.3%	82.7%	52.5%	88.5%	59.9%

Table 3: Gradient boosting accuracy ($\pm$ std): $H_0 + H_1$ landscapes alone vs. $H_0 + H_1$ landscapes combined with coordinate features.

Task	H0+H1	H0+H1+Coord
CD8 vs FoxP3	75.9% $\pm$ 9.0%	91.7% $\pm$ 6.3%
CD8 vs CD68	66.0% $\pm$ 10.8%	94.4% $\pm$ 5.7%
FoxP3 vs CD68	77.7% $\pm$ 12.7%	91.5% $\pm$ 5.9%
3-way	60.7% $\pm$ 10.1%	88.8% $\pm$ 8.8%

The coordinate-based geometric features substantially outperform landscape features across all tasks. The best coordinate models (gradient boosting) reach 91.5%–95.4% on pairwise

tasks and 88.5% on the 3-way task. Combining landscapes with coordinates yields essentially the same performance as coordinates alone (e.g. 60.7% $\rightarrow$ 88.8% on the 3-way task, versus 88.5% with coordinates only), suggesting that the coordinate features capture the necessary geometric variance to classify cells. However, it is crucial to note that this is not a naive baseline, but a highly competitive geometric representation containing 15 sophisticated spatial features (e.g., pairwise distance quantiles, convex hull area, anisotropy). Therefore, this result does not invalidate TDA’s potential, but rather suggests that standard 1-parameter TDA landscapes do not add orthogonal information to already highly dense geometric feature sets.

This result must be interpreted cautiously. It does not show that persistent homology is generally uninformative, only that in this specific setting (this dataset, this one-parameter filtration, these classifiers, this evaluation protocol), the 15 coordinate-based geometric statistics capture substantial predictive information. As discussed in Section 4.6, the coordinate baseline directly encodes nearest-neighbour and pairwise distance distributions, which implicitly contain much of the same information as one-parameter persistent homology. Richer topological representations, such as the multiparameter pipeline of Vipond et al. (2021), could capture complementary structure not recovered here.

As a robustness check for the small number of independent tumors, we repeated the main 3-way Gradient Boosting comparison using leave-one-tumor-out cross-validation. This protocol leaves out each of the 16 tumors in turn and trains on the remaining 15 tumors. The results were qualitatively consistent with the 10-fold StratifiedGroupKFold evaluation: coordinate-based geometric features achieved $86.6\% \pm 11.9\%$ mean accuracy, whereas $H_0 + H_1$ landscape features achieved $64.7\% \pm 14.4\%$. The ranking between the two feature representations is therefore stable, but the large standard deviations confirm substantial tumor-to-tumor heterogeneity.

5.4 Statistical Tests for Topological Differences

Table 4 reports the primary two-sample permutation tests on tumor-level mean H_1 landscape vectors, with Bonferroni correction ($\alpha^* = 0.0167$). Table 5 reports the paired within-tumor MMD test.

Table 4: Paired within-tumor permutation tests on tumor-level mean H_1 landscape vectors (10,000 sign-flip permutations, Bonferroni-corrected $\alpha^* = 0.0167$).

Comparison	L^2 distance	p -value	Significant?
CD8 vs FoxP3	29.84	0.1153	No
CD8 vs CD68	49.66	0.0001	Yes
CD68 vs FoxP3	52.69	0.0004	Yes

Table 5: Distributional tests on tumor-level mean H_1 landscapes. The MMD test uses 10,000 within-tumor sign-flip permutations (Bonferroni-corrected $\alpha^* = 0.0167$).

Comparison	MMD ²	p_{MMD}	Sig.?
CD8 vs FoxP3	0.1242	0.4468	No
CD8 vs CD68	0.1522	0.0116	Yes
CD68 vs FoxP3	0.1804	0.0023	Yes

The two tests yield consistent conclusions, which is itself reassuring. Both tests reject the null for the two CD68-containing comparisons (mean test: $p = 0.0001$, $p = 0.0004$; MMD: $p = 0.0116$, $p = 0.0023$) and both fail to reject for CD8 vs FoxP3 (mean: $p = 0.1153$; MMD: $p = 0.4468$). The interpretation is that macrophages (CD68) have a distinct loop-scale spatial organisation from both T cell populations, both in their mean landscape and in the full distribution of tumor-level landscapes; whereas the two T cell populations CD8 and FoxP3 do not exhibit detectable differences in their tumor-level H_1 landscape distributions at the sample size available ($N = 16$ tumors per cell type). The non-rejection should not be over-interpreted: as discussed in Section 4, it is properly read as “no evidence of difference at the tumor scale with $N = 16$ tumors per group”.

5.5 Interpretation of Mean Landscape Curves

Inspection of the mean H_1 landscape curves by cell type (Figure 10) supports the conclusions of the tumor-level tests. All three cell types exhibit their strongest topological activity at small to intermediate spatial scales, but the CD68 landscape is systematically shifted upward relative to CD8 and FoxP3, especially at the first landscape depths. This suggests that CD68 point clouds contain more pronounced loop-scale structures in the one-parameter filtration. In contrast, the CD8 and FoxP3 mean landscapes overlap substantially, consistently with the non-rejection of both the mean-permutation and MMD tests for this pair. A possible biological interpretation is that macrophages occupy spatial niches that generate more persistent loop-like structures than T cells; however, this remains hypothesis-generating, since the ROI dataset does not include direct hypoxia, necrosis, or vasculature annotations.

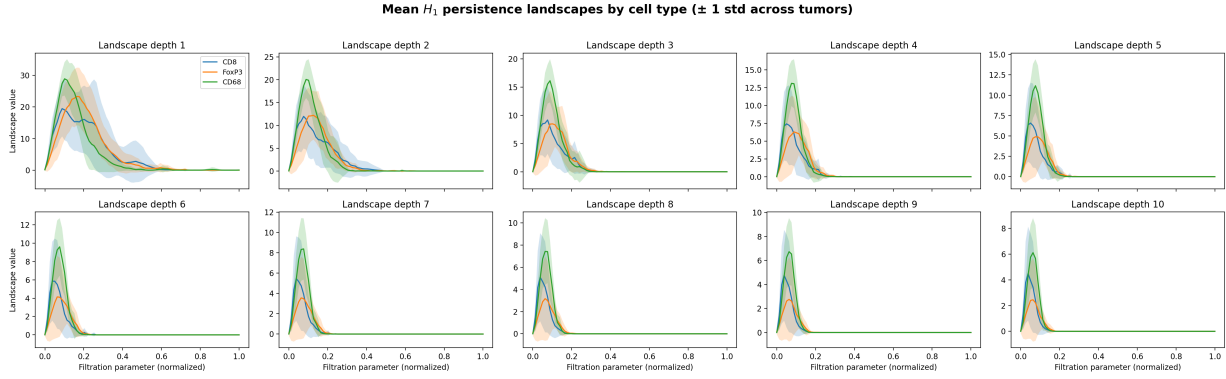


Figure 10: Mean H_1 persistence landscapes by cell type (± 1 standard deviation across the 16 tumors), for the first 10 landscape depths.

6 Conclusion

This project set out to determine whether one-parameter topological data analysis can distinguish immune cell types in the tumor microenvironment based solely on their spatial distribution. We built a complete pipeline from raw cell coordinates to topological vectorisation, classification and hypothesis testing, and evaluated it on HNSCC immunohistochemistry data from [Vipond et al. \(2021\)](#).

Our main findings are threefold. First, **one-parameter persistent homology captures statistically detectable differences between some immune cell types**. The mean-permutation tests demonstrate statistically significant differences between macrophages (CD68) and both T cell populations ($p = 0.0001$ vs CD8; $p = 0.0004$ vs FoxP3) in their H_1 landscape signatures, and the paired within-tumor MMD test confirms these results on the full distribution of tumor-level landscapes ($p = 0.0116$ and $p = 0.0023$). The classification results confirm that topological features alone achieve accuracies well above chance, particularly for tasks involving FoxP3, with gradient boosting reaching 77.1% on CD8 vs FoxP3 and 78.3% on FoxP3 vs CD68.

Second, **one-parameter persistence landscapes are outperformed by coordinate-based geometric summaries on this dataset**. These summaries are easier to compute and interpret than persistence landscapes, but they are not naive: they directly encode spread, anisotropy, nearest-neighbour distances, pairwise distances and convex-hull geometry. The 15-dimensional coordinate feature vector achieves 88.5% 3-way accuracy with gradient boosting, compared to 60.7% for $H_0 + H_1$ landscapes. Combining landscapes with coordinate features yields essentially the same performance as coordinates alone, indicating that these coordinate-based geometric features capture most of the predictive signal exploited by one-parameter persistent homology in this setting.

Third, **the two T cell populations are topologically similar at the tumor scale**.

While CD8 and FoxP3 can be separated above chance by classification (up to 77.1% with gradient boosting on H_1 landscapes), neither the mean-landscape permutation test nor the paired MMD test detects a difference at the tumor level ($p = 0.1153$ and $p = 0.4468$). The discriminative signal exploited by the classifier therefore resides in ROI-scale distributional features that are washed out by tumor-level aggregation, or in higher-order interactions among landscape coefficients that the linear permutation statistic cannot detect.

These results contextualise the findings of [Vipond et al. \(2021\)](#), who achieve substantially higher accuracies (up to 87% pairwise) using **multiparameter** persistent homology with a codensity parameter. The codensity acts as a proxy for local cell density and implicitly captures hypoxia-related spatial information that is invisible to a one-parameter Alpha complex filtration. Our work suggests that this additional parameter, or more generally the multiparameter construction, may be important for achieving the stronger performance reported by [Vipond et al. \(2021\)](#). The gap between our one-parameter results and theirs is therefore consistent with a marginal contribution of the codensity filtration, while our coordinate-versus-landscape comparison further suggests that on the standard one-parameter filtration alone, explicit coordinate-based geometric summaries already capture much of the available predictive signal.

Overall, this project demonstrates both the promise and the current limitations of one-parameter TDA for spatial biology. Persistent homology provides a principled, multi-scale summary of spatial structure that complements traditional descriptors, but in this application the one-parameter filtration studied here appears insufficient to outperform well-chosen coordinate-based geometric summaries, which already capture spread and distance-structure information directly.

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A Exact Formulation of Coordinate-Based Features

To ensure a rigorous and reproducible comparison between persistent homology and explicit spatial statistics (Section 4.6), we define the 15 coordinate-based geometric features here.

For a given ROI, let $\mathcal{P}_{raw} = \{p_1, \dots, p_n\}$ be the raw point cloud where $p_i = (x_i, y_i)$. We first center and scale the point cloud to unit Root Mean Square (RMS) radius to remove scale dependence:

$$\mathcal{P} = \left\{ \tilde{p}_i = \frac{p_i - \mu_{raw}}{R_{rms}} \right\}_{i=1}^n$$

where $\mu_{raw} = \frac{1}{n} \sum_{i=1}^n p_i$ and $R_{rms} = \sqrt{\frac{1}{n} \sum_{i=1}^n \|p_i - \mu_{raw}\|^2}$. From the normalized point cloud $\mathcal{P} = \{(\tilde{x}_i, \tilde{y}_i)\}_{i=1}^n$, we compute the following 15 features:

1. Spread and Anisotropy (4 features):

- Standard deviations along axes: $\sigma_x = \text{std}(\tilde{x})$, $\sigma_y = \text{std}(\tilde{y})$
- Axis ranges: $\Delta x = \max(\tilde{x}) - \min(\tilde{x})$, $\Delta y = \max(\tilde{y}) - \min(\tilde{y})$

2. Nearest-Neighbour (NN) Distances (5 features):

Let $d_i^{NN} = \min_{j \neq i} \|\tilde{p}_i - \tilde{p}_j\|_2$. We extract five summary statistics from the distribution of d^{NN} :

- Mean, standard deviation, minimum, maximum, and median of $\{d_i^{NN}\}_{i=1}^n$.

3. Pairwise Distances (5 features):

Let $D_{ij} = \|\tilde{p}_i - \tilde{p}_j\|_2$ for $1 \leq i < j \leq n$. We extract five summary statistics from this distribution:

- Mean, standard deviation, median, 25th percentile (Q_{25}), and 75th percentile (Q_{75}).

4. Convex Hull Area (1 feature):

- The area of the convex hull $\text{Conv}(\mathcal{P})$. In spatial point pattern analysis, the convex hull approximates the global spatial support of the point cloud. This is the smallest convex set in which the points are included.

B Software Environment

All computations were run using Python 3.13. The implementation combines topological data analysis libraries, standard scientific-computing tools, machine-learning utilities, and plotting libraries.

- **Core language:** Python 3.13
- **Topological Data Analysis:**
 - **gudhi:** construction of Alpha complexes, persistent homology computations, and persistence landscape vectorisation through `gudhi.representations.Landscape`.
 - **persim:** persistence images, persistence landscapes, and visualisation tools for exploratory topological analysis.
- **Scientific computing and data manipulation:**
 - **numpy:** numerical arrays, vectorised operations, random number generation, and linear-algebra utilities.
 - **pandas:** loading, cleaning, grouping, and aggregating ROI-level and tumor-level data.
 - **scipy:** pairwise distances, convex hull computations, and auxiliary statistical tools.
- **Machine learning and statistical modelling:**
 - **scikit-learn:** preprocessing, pipelines, cross-validation, classifiers, model selection, metrics, nearest-neighbour computations, and RBF kernels. In particular, we use `StandardScaler`, `Pipeline`, `StratifiedGroupKFold`, `LeaveOneGroupOut`, `LinearDiscriminantAnalysis`, `LogisticRegression`, `GradientBoostingClassifier`, `GridSearchCV`, `confusion_matrix`, `accuracy_score`, `NearestNeighbors`, and `rbf_kernel`.
 - **joblib:** parallel computation of persistence diagrams and feature extraction routines.
- **Visualisation:**
 - **matplotlib:** figures, persistence diagrams, landscapes, confusion matrices, and result visualisations.
 - **seaborn:** statistical plots and improved visual presentation of exploratory analyses.

- **Standard-library modules:**

- `os`, `glob`, and `pathlib`: file-system navigation and dataset loading.
- `itertools`: construction of pairwise cell-type comparisons and iteration utilities.

C Statistical Testing Framework

This appendix formalises the two statistical tests used to compare the spatial organisation of two immune cell types. Throughout this appendix, A and B denote two cell types among CD8, CD68 and FoxP3.

For each tumor $t \in \{1, \dots, T\}$ and each cell type $c \in \{A, B\}$, we first compute the tumor-level mean H_1 persistence landscape vector:

$$\bar{\lambda}_{t,c} = \frac{1}{n_{t,c}} \sum_{i=1}^{n_{t,c}} \lambda_{t,c,i}^{(H_1)} \in \mathbb{R}^d,$$

where $\lambda_{t,c,i}^{(H_1)}$ is the H_1 landscape vector of ROI i from tumor t and cell type c . In our implementation, $d = 800$, corresponding to 10 landscape depths evaluated on 80 grid points.

For a fixed pair of cell types (A, B) , define

$$X_t = \bar{\lambda}_{t,A}, \quad Y_t = \bar{\lambda}_{t,B}.$$

Thus, each tumor provides one paired observation (X_t, Y_t) .

C.1 Test 1: equality of mean topological signatures

Statistical question. The first test asks whether the two cell types have the same average tumor-level topological signature. In other words, it tests whether the expected tumor-level mean H_1 landscape vector is the same for cell types A and B .

Hypotheses. Let

$$\mu_A = \mathbb{E}[X_t], \quad \mu_B = \mathbb{E}[Y_t].$$

The statistical hypotheses are

$$H_0 : \mu_A = \mu_B$$

against

$$H_1 : \mu_A \neq \mu_B.$$

Equivalently,

$$H_0 : \mathbb{E}[X_t - Y_t] = 0$$

against

$$H_1 : \mathbb{E}[X_t - Y_t] \neq 0.$$

This is a test on the mean of the tumor-level landscape vectors. It does not test equality of the full distributions.

Test statistic. We use the Euclidean distance between the two empirical mean landscape vectors:

$$T_{\text{obs}} = \left\| \frac{1}{T} \sum_{t=1}^T X_t - \frac{1}{T} \sum_{t=1}^T Y_t \right\|_2.$$

Equivalently, writing $D_t = X_t - Y_t$,

$$T_{\text{obs}} = \left\| \frac{1}{T} \sum_{t=1}^T D_t \right\|_2.$$

Large values of T_{obs} indicate evidence against H_0 .

Permutation scheme. Because X_t and Y_t are measured within the same tumor, the test must preserve the paired structure. Under H_0 , the two labels A and B are exchangeable within each tumor. Therefore, for each permutation $b = 1, \dots, B$, we draw independent signs

$$\varepsilon_t^{(b)} \in \{-1, +1\}, \quad \mathbb{P}(\varepsilon_t^{(b)} = 1) = \mathbb{P}(\varepsilon_t^{(b)} = -1) = \frac{1}{2},$$

and compute

$$T_b = \left\| \frac{1}{T} \sum_{t=1}^T \varepsilon_t^{(b)} D_t \right\|_2.$$

Rejection region. This is an upper-tail test. At level α , the rejection region is

$$\mathcal{R}_\alpha = \{T_{\text{obs}} \geq q_{1-\alpha}\},$$

where $q_{1-\alpha}$ is the empirical $(1 - \alpha)$ -quantile of the permutation statistics $(T_b)_{b=1}^B$.

Equivalently, we reject H_0 when

$$\hat{p} \leq \alpha.$$

Empirical p-value. With $B = 10,000$ permutations, the empirical p-value is

$$\hat{p} = \frac{1 + \sum_{b=1}^B \mathbf{1}\{T_b \geq T_{\text{obs}}\}}{B + 1}.$$

C.2 Test 2: equality of distributions of topological signatures

Statistical question. The second test asks whether the full tumor-level distributions of H_1 landscape vectors are the same for the two cell types. This is stronger than the previous test: two cell types may have the same mean landscape but different variance, dispersion, or distributional shape.

Hypotheses. Let P_A be the distribution of X_t , and let P_B be the distribution of Y_t :

$$X_t \sim P_A, \quad Y_t \sim P_B.$$

The statistical hypotheses are

$$H_0 : P_A = P_B$$

against

$$H_1 : P_A \neq P_B.$$

This is a two-sample distributional test on tumor-level landscape vectors.

Kernel. We use the Gaussian radial basis function kernel

$$k(x, y) = \exp \left(-\gamma \|x - y\|_2^2 \right),$$

where $\gamma > 0$.

Test statistic. The test statistic is the empirical squared Maximum Mean Discrepancy. The implementation uses the biased V-statistic estimator:

$$M_{\text{obs}} = \widehat{\text{MMD}}_b^2(X, Y),$$

where

$$\widehat{\text{MMD}}_b^2(X, Y) = \frac{1}{T^2} \sum_{i=1}^T \sum_{j=1}^T k(X_i, X_j) + \frac{1}{T^2} \sum_{i=1}^T \sum_{j=1}^T k(Y_i, Y_j) - \frac{2}{T^2} \sum_{i=1}^T \sum_{j=1}^T k(X_i, Y_j).$$

Large values of M_{obs} indicate evidence against H_0 .

Permutation scheme. Because observations are paired by tumor, permutations are performed within each tumor. For each permutation b , define

$$(X_t^{(b)}, Y_t^{(b)}) = \begin{cases} (X_t, Y_t), & \text{if } \varepsilon_t^{(b)} = 1, \\ (Y_t, X_t), & \text{if } \varepsilon_t^{(b)} = -1. \end{cases}$$

Then compute

$$M_b = \widehat{\text{MMD}}_b^2(X^{(b)}, Y^{(b)}).$$

Rejection region. This is an upper-tail test. At level α , the rejection region is

$$\mathcal{R}_\alpha^{\text{MMD}} = \{M_{\text{obs}} \geq q_{1-\alpha}^{\text{MMD}}\},$$

where $q_{1-\alpha}^{\text{MMD}}$ is the empirical $(1 - \alpha)$ -quantile of the permuted MMD statistics.

Equivalently, we reject H_0 when

$$\hat{p}_{\text{MMD}} \leq \alpha.$$

Empirical p-value.

$$\hat{p}_{\text{MMD}} = \frac{1 + \sum_{b=1}^B \mathbf{1}\{M_b \geq M_{\text{obs}}\}}{B + 1}.$$

C.3 Multiple testing correction

For each test, we perform the three pairwise comparisons

$$\text{CD8 vs FoxP3}, \quad \text{CD8 vs CD68}, \quad \text{CD68 vs FoxP3}.$$

To control the family-wise error rate at level 5%, we use the Bonferroni correction:

$$\alpha^* = \frac{0.05}{3} \approx 0.0167.$$

Therefore, for each pairwise comparison and each test, the final decision rule is

$$\text{Reject } H_0 \iff \hat{p} \leq 0.0167.$$